

Applied Mathematics and Informatics in Drug Discovery (AMIDD)

Course outline - Fall semester 2026, University of Basel

Drug discovery is an interdisciplinary craft. In the AMIDD course, we will get an impression of how researchers in drug discovery and development use mathematics as a language and computation as a tool to understand biology and drug-body interactions.

The course's website, AMIDD.ch, contains all essential information, including the syllabus and course material (slides, papers to read, submission links for offline activities). If you have questions, you can reach Jitao David Zhang, the lecturer, at jitao-david.zhang@unibas.ch.

What we will learn together

We start the journey with real-world examples of clinically approved drugs such as semaglutide (brand names *Ozempic* and *Rybelsus* for diabetes, and *Wegovy* for weight management). We back-trace the discovery and development of the medicine through the linear model of the drug-discovery pipeline, and explore the interaction between semaglutide and the human body by adopting a multiscale view across molecular, cellular, organ and system, and population levels. From there, we will explore the major players and stakeholders of the drug discovery process (lecture 2), and learn the key questions (lecture 3).

Having built the foundation, we go on to study the biological foundations of drug discovery (lecture 4), in particular proteins, the targets of most drugs (lecture 5). We will examine how statistical, machine learning and artificial intelligence models (lecture 6), as well as causal models (lecture 7), are applied.

Examples of models and their applications are introduced in the context of lead identification and optimization (lecture 8), as well as understanding the mechanism and mode of action of drugs (lecture 9). Finally, we use PK/PD modeling as an example of ODE-based models and examine the basics of clinical trials (lecture 10). The course ends with guest lectures (lecture 11) and a collaboration challenge (lecture 12), where participants form teams to achieve a common goal within limited time and resources.

Agenda

#	Date	Session	Scope
1	18.09.	Introduction to drug discovery	The whole arc, seen through one drug
2	25.09.	The <i>What</i> , the <i>Who</i> , and the <i>How</i>	Workflow, stakeholders, paths to a drug
3	02.10.	Key questions in drug discovery	The five questions as a project compass
4	09.10.	Biological foundation of drug discovery	From genes to phenotypes
5	16.10.	Protein as drug target	Molecular, physics-based models
6	23.10.	Statistical, machine learning, and artificial intelligence models	Statistical modeling
7	30.10.	Causal inference	Causal models, generative simulation
-	06.11.	No lecture - excise planned	
8	13.11.	Lead identification and optimization	Molecule-level design and profiling
9	20.11.	Mechanism and mode of action	Omics, networks, cell-level models
-	27.11.	<i>Dies academicus</i> — no lecture	
10	04.12.	PK/PD modeling and clinical trials	Organ, organism, and population models
11	11.12.	Guest lectures	Practitioners' perspectives
12	18.12.	A collaboration challenge	Everything above, applied in teams

Topics we will cover

Drug discovery and biomedicine: Modalities (small molecules, peptides, antibodies, oligonucleotides, and emerging ones); medical need and indication selection; target identification, assessment, and validation; the biological basis of disease; proteins as drug targets and structure-function relationships; lead identification and optimization; ADME (absorption, distribution, metabolism, and excretion), potency, selectivity, and safety profiling; mechanism and mode of action; pharmacokinetics and pharmacodynamics; clinical trials; regulatory milestones and relevant stakeholders.

Mathematics and informatics. Mechanistic and physics-based modelling; binding kinetics and dose-response; statistical inference; machine learning, transformer-based AI models, and their evaluation; structure prediction and ligand-based design; causal inference; sequence analysis and language models; omics data analysis and network biology; multiscale modelling; PK/PD and physiologically based (PBPK) models; population models; reproducibility and the responsible use of LLMs and AI agents.

How we work, and how the grades are given

An ideal classroom is diverse, interactive, and curious: everyone brings expertise the rest of us lack. Each lecture is followed by one or more offline activities, such as reading a paper, watching a video, or trying out an analysis. Sometimes I will send out a short, anonymous survey to collect your feedback.

Your grade is given by participation including quizzes, offline activities, and the collaboration challenge.